FIVE

Organizing the Manned Space Program

> The really significant fallout from the strains, traumas, and endless experimentation of Project Apollo has been of a sociological rather than a technological nature; techniques for directing the massed scores of thousands of minds in a close-knit, mutually enhancive combination of government, university, and private industry.
>
> —T. Alexander, in *Fortune*

By far the largest programs within the National Aeronautics and Space Administration (NASA) during the 1960s were the manned space projects Mercury, Gemini, and Apollo. These differed from other NASA programs because of their massive scale and because several field centers, not just one, contributed significantly to them. The NASA headquarters role was bigger for these huge projects than it was for smaller ones: headquarters coordinated the work of the different field centers. The manned space program contributed disproportionately to the management philosophy and style of NASA as a whole, defined by agency-wide procedures.[1]

While astronauts grabbed public attention, NASA managers and engineers quietly created the machines and procedures necessary for astronauts and ground controllers to operate them. With their personnel descended from German rocket pioneers and National Advisory Committee for Aeronautics (NACA) researchers, NASA's informal groups brought years of aircraft and rocket design expertise to spacecraft design. These new technologies demanded strict attention, and there were the usual number of failures. NASA personnel had to learn how to design manned spacecraft and man-rated rockets as well as how to direct thousands of new employees and scores of contractors.

Difficulties in making the transition from engineers to managers led NASA executives to look elsewhere for people with strong organizational skills. Executives turned primarily to the air force, an organization that developed technologies similar to NASA's. From its inception, NASA had used military personnel, but the importation of experienced air force officers reached its peak in 1964 and 1965, as the newly installed Apollo program director, Brig. Gen. Samuel C. Phillips of the air force, arranged the transfer of scores of air force officers to bring order to NASA's chaotic committees. Phillips imported air force methods such as configuration control, the Program Evaluation and Review Technique (PERT), project management, and Resident Program Offices at contractor locations. By the end of Apollo, Phillips had grafted significant elements of Air Force Systems Command (AFSC) onto NASA's original culture.

Management by Committee, 1958–1962

At its inception in October 1958, NASA consisted of field centers transferred from other organizations. Three centers from NACA formed NASA's core: the Langley Aeronautical Laboratory in Hampton, Virginia; Lewis Research Laboratory in Cleveland, Ohio; and Ames Research Laboratory near Sunnyvale, California. NACA researchers concentrated on empirical and mathematical investigations of aircraft design, including the "X series" of high-performance aircraft, high-speed aerodynamics, jet engines, and rocket propulsion.[2]

An ad hoc group of NACA researchers known as the Space Task Group (STG) promoted the development of space flight. By 1958, they had developed a blunt body capsule design to put a man into space. One other organization, transferred to NASA in January 1960, was key to NASA's manned space efforts: Wernher von Braun's Army Ballistic Missile Agency (ABMA) in Huntsville, Alabama, and its launch facilities at Cape Canaveral, Florida. NASA renamed the ABMA the Marshall Space Flight Center (MSFC), and the Cape Canaveral facilities eventually became the Kennedy Space Center (KSC).[3] These two centers created the massive rockets and launch facilities necessary to place men on the Moon.

The STG's manned capsule, now christened *Mercury*, topped NASA's

agenda. Engineers in the Mercury project were to create not only a space capsule but also a worldwide communications network. They were to marry the capsule and the network to the launchers under development by the army and air force. Langley Assistant Director Robert Gilruth headed the STG, which grew quickly and in 1962 moved to Houston, Texas, becoming the Manned Spacecraft Center (MSC).[4]

Gilruth was typical of NASA's experienced engineering researchers. He graduated in 1936 with a master's degree in aeronautical engineering from the University of Minnesota, where he designed high-speed aircraft. From Minneapolis he moved to Langley Research Laboratory, developing quantitative measures for aircraft flying qualities, a job that later served him well in developing manned spacecraft. His *Requirements for Satisfactory Flying Qualities of an Aircraft* became the standard for the field for some years. Gilruth's next assignment brought him back to high-speed aircraft: developing wind tunnel techniques to measure hypersonic flow. Hypersonic flow problems led him to perform full-scale experiments dropping objects from high altitudes at Wallops Island off the Virginia coast. These experiments brought him into contact with rocketry, as he and his NACA colleagues developed and launched rockets to test their theories and technologies.[5]

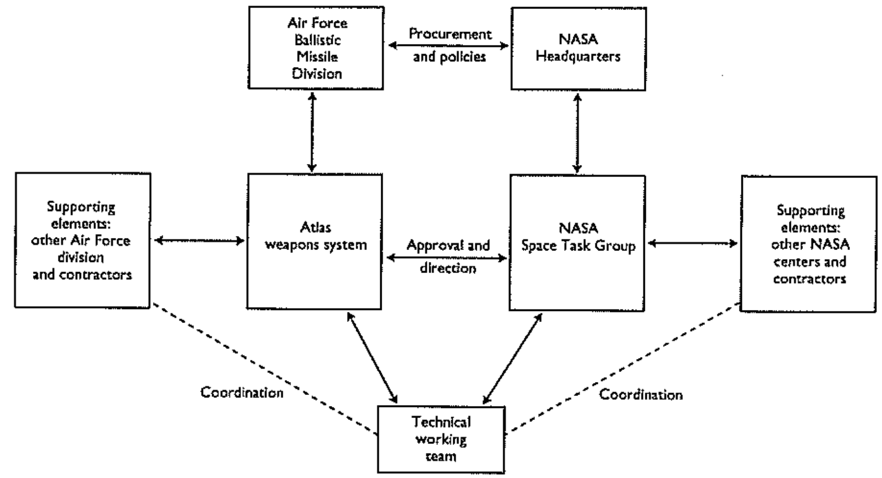
In the early manned programs, Gilruth treated his engineering colleagues as technical equals. As his assistant, Paul Purser, described it, "Individuals around the conference table are not aware of being division chiefs or section heads—they are all people working on a problem." Gilruth's ability and experience made him more than just a manager. Future NASA Administrator George Low said in the early 1960s, "Gilruth works personally with many people in the Space Task Group. His method of operation is one of very close technical involvement in the project. He could tell you . . . every nut and bolt in the Mercury capsule, how it works, and why it works. I've been in many meetings over the last two or three years, where the whole picture would look very complex. After perhaps a half-hour's discussion, Gilruth would come up with the right solution, and the rest of us present would wonder why we hadn't thought of it."[6] The STG's hands-on approach to engineering would continue for years to come.[7]

In the fall of 1958, the STG established Mercury's basic configurations and missions. Engineers planned to use existing rockets to launch the new spacecraft—first von Braun's proven *Redstone* and *Jupiter* boosters, then the air force's more powerful but less mature *Atlas*. Congress gave NASA the same procurement regulations as the Department of Defense (DOD). Thus the new organization held a bidder's conference in November 1958 to describe the proposed system to contractors, mailed out bid specifications, and required responses in 30 days. In January 1959 NASA awarded the *Mercury* spacecraft contract to McDonnell Aircraft Corporation, a cost-plus-fixed-fee contract for $18,300,000 and an award fee of $1,150,000. STG engineers also started negotiations with the army and the air force for launchers.[8]

Two traditions distinguished Mercury's management: the informal structure and procedures of the STG, and the more formal approaches of McDonnell Aircraft and the air force. STG engineers and scientists used committees characteristic of research, simply creating more of them as Mercury grew. McDonnell Aircraft brought structured methods developed from years of interaction with the air force. Engineers from the STG and McDonnell Aircraft worked closely to resolve the numerous novel problems they encountered. For Atlas, the air force used its own procedures and supplied representatives to STG committees to define interfaces between *Atlas* and the *Mercury* capsule.[9]

Mercury's driving force was a "bond of mutual purpose": determination to regain national prestige, fear of Soviet technical accomplishments, and pride in American capabilities. Managers gave tasks and the authority to perform them to young engineers. As one STG veteran put it, "NASA responsibilities were delegated to the people and they, who didn't know how to do these things, were expected to go find out how to do it and do it."[10] Working teams phased in and out as they completed their tasks; frequently they determined "a course of action and proceeded without further delay, with verification documentation following through regular channels." In other words, engineers took immediate action without management review and left others to clean up the paperwork.[11]

This worked because of the extraordinary "flatness" and open communication prevalent throughout the organization. STG leaders insisted that problems be brought into the open. According to Chris Kraft, later the director of Johnson Space Center, all the people in the organization felt as if they could say "what they wanted to say any time they wanted to say it"; Kraft called



The Mercury-Atlas organization was extraordinarily flat, with only three levels from NASA and air force headquarters to the working groups who built the spacecraft and launchers. Adapted from *Mercury Project Summary, Including Results of the Fourth Manned Orbital Flight*, SP-45 (Washington, D.C.: NASA, 1963), 19, figure 1-8.

this STG's "heritage."[12] Managers tracked events through informal communications and frequent technical reviews. They directed resources to problem areas, but they seldom intervened.[13]

On Mercury, and later on Gemini and Apollo, this sense of common purpose also prevented the development of bureaucratic sclerosis. As stated by one of the engineers on Gemini and Apollo, "You would see people who would try to build empires, who would try to be obstructionist, and they would be just absolutely steamrolled by this team. I saw it time and time again where there was this intense feeling of teamwork. It wasn't always smooth, but it was like, 'We've got a common goal.'"[14]

The STG's multiple committees became unwieldy as Mercury grew from a nucleus of 35 people in October 1958 to 350 in July 1959. To deal with the proliferation of committees, STG created another committee, the Capsule-Coordination Panel, subsequently upgraded to an office in Washington, D.C.[15]

NASA headquarters executives soon realized that the STG and McDonnell had drastically underestimated the scope, cost, and schedule of the program. Within two months of its beginning, the estimated cost of the McDonnell contract was $41 million, more than twice the initial estimate, while the air force's estimated costs for the *Atlas* boosters increased from $2.5 million to $3.3 million. These increases led to a round of cost-cutting measures, yet costs continued to rise. The estimated cost of the McDonnell contract reached $70 million by January 1960. NASA Administrator Keith Glennan's initial response was to visit the STG in May 1959. He came away impressed by the esprit de corps in the STG and the size and complexity of the project. With Congress and the administration willing to foot the bill and the STG rapidly tackling technical issues, Glennan elected not to intervene.[16]

In the summer of 1959, Gilruth organized the New Projects Panel to identify manned projects beyond Mercury. The panel identified circumlunar flight (not landing) as the most promising goal.[17] The most important technical developments for a manned Moon mission were in the military's rocket and engine programs. In January 1959, NASA acquired the air force contract with North American Aviation (NAA) to develop the huge liquid-fueled *F-1* engine. NASA acquired the *Saturn I* launcher in January 1960 along with von Braun's rocket team. *Saturn* was the only launch vehicle then under development that promised sufficient size for a manned lunar landing program. However, for the moment, it was a launch vehicle without a mission.[18]

In early 1960, NASA's advanced planning groups concluded that a lunar mission was the best next step. NASA named the proposed new program Apollo, and in August managers announced they would award three contracts to industry for feasibility studies. The STG selected the Martin Company, General Electric (GE), and the General Dynamics Convair Division to perform the studies. Study guidelines were so vague that when the Martin Company engineers reported back in December 1960, STG engineers told them to include astronauts and to consider lunar landing and recovery. MSFC also sponsored its own feasibility studies, while the STG started an internal study.[19]

After the Bay of Pigs disaster and Yuri Gagarin's flight in April 1961, the Kennedy administration proved receptive to NASA's lunar mission planning. On May 25, President John F. Kennedy proposed that NASA land a man on the Moon "before the decade is out." Congress enthusiastically agreed and immediately increased NASA's funding.[20]

STG managers quickly moved Apollo from feasibility studies to development. Gilruth had prepared the groundwork, creating the Apollo Project Office in September 1960. Although the hardware configuration remained uncertain, the STG forged ahead, dividing the system into six contracts: launch vehicles, spacecraft command module and return vehicle, propulsion module, lunar landing stage, communications and tracking network, and launch facilities. Just before selecting NAA for the spacecraft command module in November 1961, MSC engineers changed the Statement of Work, meaning that they awarded NAA a contract to build a command module based on specifications that NAA's managers and engineers had never seen.[21]

With four Saturn stages under development, von Braun's MSFC engineers did not have the resources to design, manufacture, and test all of the vehicles. They had to rely upon industry instead of their traditional in-house design. MSFC managers transferred *S-I* stage development and manufacturing to Chrysler and awarded the new *J-2* cryogenic engine to NAA Rocketdyne in June 1960.[22]

MSFC inherited strong technical divisions from its army heritage, each based on specific disciplines such as rocket propulsion, structures, or avionics. The technical divisions coordinated project work through committees, contributing to MSFC's typically large, interminable meetings. Contractors complained that MSFC managed by technical takeover. However, under the pressure of having many large, complex projects, project and matrix management made inroads into MSFC's discipline-based, functional organization. The divisions fought the change, forcing MSFC Director Wernher von Braun to clarify the power of project managers vis-à-vis the technical divisions.[23]

Von Braun was one of the world's leading rocket engineers and a charismatic leader. Even when immersed in administrative duties, he closely followed the technical details of MSFC's rockets. Von Braun secured inputs from all participating engineers and technicians and arrived at a consensus through group meetings. He used an informal but disciplined system of weekly notes, requiring subordinates *two* levels below him to send him one page of notes summarizing the week's events and issues. Von Braun then wrote comments in the margins of these notes, copied the entire week's set, and distributed the notes for everyone in MSFC to read. They became popular reading because they contained the boss's detailed comments on MSFC events and people.

This system of "Monday Notes" had a number of important ramifications. First, because von Braun required the notes to come from two levels below, the managers directly under him could not edit the news he received. Second, because of having to send weekly notes to von Braun, all managers formed their own information-gathering mechanisms. Third, the redistributed notes with von Braun's marginalia provided a mechanism for cross-division information flow, because everyone saw comments on not only their own activities but the activities of all other divisions. The notes moved information vertically—from the managers and engineers up to von Braun, and from von Braun down to the managers and engineers—as well as horizontally—from division to division.

This very open communication system provided MSFC engineers and managers with advanced notice of potential problems, often spurring critical problem-solving efforts across the divisions. Some MSFC engineers complained about the extraordinary communication technique because it created an "almost iron-like discipline of organizational communication" in which "nobody at the bottom really felt free to do anything unless he got it approved from the next level up, the next level up, the next level up." However, it did ensure that information flowed quickly and effectively throughout the organization.[24]

Von Braun required all MSFC personnel to take "automatic responsibility" for problems. If MSFC employees found a problem, they were to solve it, find someone who could solve it, or bring it to management's attention, whether or not the problem was in their normal area of responsibility. This intentional blurring of organizational lines helped create an organization more interested in solving problems than in fighting for bureaucratic turf.[25]

Apollo planners soon recognized a gap between Mercury's short flights and the long flights and complex operations of Apollo. To bridge this gap, STG chief Gilruth authorized the Gemini program, which was to modify *Mercury* to accommodate two astronauts, perform orbital maneuvers, and rendezvous with other spacecraft. NASA awarded the *Gemini* capsule contract to McDonnell without competition because it was a modification of *Mercury*. Gilruth split the engineering staff between Mercury and Gemini, and in January 1962 he established the Mercury, Gemini, and Apollo Project Offices.[26]

Like Mercury and Apollo, Gemini used coordination panels for day-to-day management. The Project Office established six panels: three for the spacecraft—mechanical systems, electrical systems, and flight operations—and one each for the paraglider,[27] *Atlas-Agena*, and *Titan II*. They typically held weekly

meetings, while the air force used its standard procedures to manage its portions of the program. Because the air force provided the *Titan II* and the *Agena* target boosted on an *Atlas* missile, air force and NASA managers established an additional panel to coordinate between them. Assistant Secretary of the Air Force for Research and Development Brockway McMillan and NASA's Robert Seamans were the co-chairmen, with D. Brainerd Holmes of the Office of Manned Space Flight (OMSF) and Gen. Bernard Schriever of AFSC the highest-ranking members.[28]

Committees coordinated between engineers and managers at headquarters, MSFC, and MSC, particularly for interface designs and characteristics. By July 1963, there were so many committees that Holmes created another one, the Panel Review Board, to coordinate them.[29]

In the white heat of the early post-*Sputnik* era, technical achievement was the primary gauge of space program success, and political leaders left control in the hands of the engineers who promised technical success. Engineers and scientists from the STG and MSFC used committees to coordinate their work, a habit inherited from research traditions of NACA laboratories and von Braun's "Rocket Team." Engineers rapidly developed rockets and spacecraft, with little heed for cost. NASA and its contractors rushed into contracts and designs without firm requirements or a clearly defined mission, making schedule or cost predictions virtually impossible. For example, MSC and MSFC engineers wrote definitive specifications for their Apollo elements well after contract awards—and in the case of MSFC's *Saturn* stage I, long after completion of the initial design, manufacturing, and testing.[30] For the moment, this was not a problem, because Congress gave NASA more money than NASA asked for, allowing a continuation of conservative design traditions on a much larger scale.

Conservative Engineering and the State of the Art

MSFC's conservative tradition of rocket development traced back to von Braun's work with the German Army in World War II and the U.S. Army thereafter. Step-by-step, precise methods characterized von Braun's approach to rocketry. Each rocket drew upon the designs and experiences of previous rockets. Engineers made small changes in each successive rocket and tested each change to ensure that it did not compromise the design. They considered flights successful even if they ended in explosion as long as they collected sufficient data to determine the explosion's cause.[31] Once designers found the problem, they modified the design and flew it to test the fix. Based on their many years of experience, they believed it virtually impossible to design an engine or a rocket without significant difficulties.

Although MSC and MSFC engineers differed in a number of ways, one similarity was distrust of contractors. Essentially, both MSFC and MSC engineers took apart and rebuilt the stages and capsules that contractors delivered to them. Boeing's contract for the *Saturn S-I* first stage was a good example. MSFC's original contract, awarded after a competition in 1962, gave Boeing partial responsibility for stage design and assembly but little responsibility for booster specifications or testing. Later contracts gave Boeing progressively more responsibility, until it was responsible for all aspects of the stage except for mission operations.[32]

Borrowing and extending practices from the air force and the navy, NASA closely controlled the contractors. NASA used Resident Manager's Offices to monitor the contractors. NASA realized that on-site surveillance was "somewhat sensitive from the point of view of the contractor" but persisted because of its belief in face-to-face communication, its distrust of contractor capabilities, and its trust in its own capacities. NASA's infringement on contractor prerogatives included forced renegotiation of contracts and designs. For example, Lunar Excursion Module (LEM) contract winner Grumman supposed that it would build the design presented in its bid. Instead, NASA engineers redesigned the LEM and the contract.[33]

NASA Administrator James Webb expanded contractor penetration to penetration of its own field centers. He wanted a separate information channel to check his own organization, and he hired GE for this purpose. The GE Policy Review Board, established in December 1962, was to provide systemwide coordination and integration. Neither MSC personnel nor MSFC personnel wanted headquarters or GE to integrate them, however. As one GE manager later explained, "Frankly, they didn't want us. There were two things against us down there [at MSC]. No. 1, it was a Headquarters contract, and it was decreed that the Centers shall use GE for certain things; and [No. 2] they considered us Headquarters spies."[34] MSC management hampered GE's attempts to fulfill its integration role, prohibiting "unannounced visits" and forbidding GE from taking any significant action unless approved by MSC. Field center resistance was effective, for in July 1963, Apollo's new Panel Review Board abolished GE's board. Later, after a long briefing to Administrator Webb, Apollo program director Phillips removed systems integration from GE's contract.[35]

Headquarters also contracted with American Telephone and Telegraph (AT&T) to provide technical assistance. AT&T created a separate company called Bellcomm to provide headquarters with technical consultants, whom headquarters used to cross-check the field centers in the way that The Aerospace Corporation checked contractors for the air force. Bellcomm ultimately performed exemplary work in trajectory analysis, but field center engineers and managers avoided Bellcomm representatives because they believed them to be headquarters spies. The use of GE and Bellcomm failed to provide NASA executives with the means to control the field centers, because of the unequal power between the government and industry. NASA provided the funds, and industrial contractors were uncomfortable criticizing their source of funding. They knew better than to bite the hand that fed them.[36]

Perhaps the greatest challenge to NASA and contractor engineers was to make their new vehicles absolutely reliable so that humans could fly in them. Redundancy was one common technique to improve reliability. For example, MSFC engineers designed each *Saturn* stage with clusters of engines, so that if one failed, the remaining engines could continue the mission. Man-rating the air force's *Atlas* and *Titan* missiles primarily meant adding redundant electronic circuitry and failure detection circuits to the missile. NASA engineers thought of astronauts as redundant design elements that could improve the chances for mission success by taking over spacecraft functions if components failed.[37]

Ensuring high quality and safety was as much a function of management and organization as engineering design. Because a number of booster and capsule components were single string—meaning that if they failed, there were no separate backups—STG engineers rigorously verified individual components for high quality. In turn, quality depended upon every worker building each component with the finest workmanship.

The manned programs instituted special methods to make factory workers aware of the importance of high quality. Taking advantage of the projects' prestige and visibility, Mercury and Gemini managers distinguished NASA components and workers using special symbols. On the Gemini Titan II program, Martin management gave special worker certifications to top employees, along with orientations, emblems, labels, badges, and even distinctive toolboxes, "painted Air Force blue and individualized with each worker's name." Astronauts visited production lines to encourage high standards and workmanship. Based upon Space Technology Laboratories (STL) recommendations, the program adopted strict inspections and random part checking for quality control. Tight control over manufacturing contrasted sharply with MSC's loose internal structure and processes.[38]

Worker motivation for Apollo was extraordinary, contributing significantly to the high quality of components that went into the project. Most of the time, this was a major advantage. However, occasionally, it caused problems. For example, at NAA, the wiring harnesses for the command module occasionally disconnected when the pins broke off in the connectors. They "found one lady out there who had evidently extremely strong hands, and she knew that she was working on the *Apollo* Program, so when she crimped, she crimped extremely hard, and she could actually crimp hard enough to deform the tool and squeeze the wires to where they were almost broken. She was just trying to do her job a little better than normal, but actually she was causing us a lot of trouble. For her, they put a spatial stop on the tool so that she couldn't crimp it any harder."[39]

Mercury and Gemini engineers created a system to track individual parts with manufacturing and test histories—to ensure that only fully documented, flawless parts became flight components. General Dynamics and Martin managers designated *Mercury* and *Gemini* launch vehicle components as critical, with special tags, paperwork, and procedures. At Martin, vehicle chaperones accompanied each vehicle through manufacturing, moving the vehicle and its paperwork through the factory. Using Martin's program as a model, the air force and NASA initiated a similar program at Lockheed for the *Agena* target vehicle. At the Factory Roll-Out Inspection, NASA thoroughly reviewed records for each vehicle component. Mercury engineers started their quality control program late, resulting in many component replacements when the

capsule or its components failed acceptance tests. Learning from this, Gemini managers began their quality control program right at the start.[40]

Military models were the basis for most of the STG's few formal processes early in the program. The Source Evaluation Board was one example, as was the Mock-Up Inspection Board, a formal inspection of a full-scale system model. Another was the Development Engineering Inspection, where STG engineers certified the design to be ready for flight. Hardware qualification was another military import, using rigorous environmental tests to stress components. On the *Apollo* spacecraft, contractors prepared for the Spacecraft Assessment Review, Customer Acceptance Readiness Review, and Flight Readiness Review. Contractors submitted written reports, which NASA committees reviewed. Starting on Mercury, NASA also held flight safety reviews for each spacecraft to review modifications, testing, and preparations for launch and operations.[41]

Engineers at both centers believed in uncovering design problems through extensive tests and analyses. STG Director Gilruth and MSFC Director von Braun both believed that testing led to better understanding and improved designs. MSFC engineers tested engine stability by exploding small bombs in the rocket's exhaust path to ensure that unexpected flow variations in the engine's hot gases would not create instabilities that could lead to an explosion. STG engineers injected electrical failures into their designs to ensure that they would survive. Following air force ballistic missile practices, they searched for critical weaknesses in the design, making changes when they found them.[42]

NASA and its contractors tested components to ensure that they could survive launch vehicle vibrations, the thermal and vacuum environment of space, and electromagnetic interference between electronic components. They also performed life tests to see how long components would last. Once components completed component tests and Preinstallation Acceptance Tests, engineers and technicians integrated them into a capsule. Contractors then ran Capsule System Tests, which verified that electronic wiring was working and that mechanical devices were functioning properly.[43]

One item not easily tested was the performance of the environmental and thermal control system of the entire *Mercury* capsule. Although vacuum chambers existed, none were large enough to test the whole vehicle. In late

1960, STG and McDonnell engineers developed a large vacuum chamber in which they could test the entire capsule's environmental and thermal controls. This task, known as Project Orbit, found numerous problems, particularly in the reaction control system used to keep the capsule pointed correctly in space.[44]

To manage the large number of engineering alterations, the STG established for Mercury a Change Control Group, whose membership fluctuated based on the nature of the problem. Configuration control crept into the program through the air force's supply of launchers for NASA. The air force established configuration control for Gemini's Titan II in December 1962. Gemini engineers "froze" portions of the spacecraft design as early as March 1962, and Apollo contractors froze elements of their designs by June 1963.[45]

Despite these efforts, *Mercury*'s first test flights suffered their share of failures. The first occurred in July 1960 during the first full-scale test of the *Mercury* capsule mounted on an *Atlas* booster. About one minute after launch, the booster failed as it accelerated through the period of highest aerodynamic pressure, leaving NASA and the air force to search for debris in the Atlantic off Cape Canaveral. The evidence pointed to a structural failure in the interface between the *Atlas* and the *Mercury* capsule, based on mechanical differences between the *Mercury* capsule and the *Atlas*'s normal complement of nuclear warheads. Engineers had not found this problem in earlier tests because they could not fully simulate the physical dynamics of the vehicle in flight. Following the failure, the Mercury-Atlas coordination panel formed a new committee to find and resolve interface problems. The next successful test flight, in February 1961, featured a structure-stiffening "belly band." Engineers later included vibration and structural resonance characteristics of the combined launch vehicle–payload system in interface designs and documentation.[46]

Interface problems also dogged a test flight aboard MSFC's *Redstone* booster in November 1960. The launcher lifted off the ground about four inches, then settled back onto the pad while the capsule's escape rocket launched, dragging erroneously opened parachutes. Subsequent investigations traced the failure to a timing problem between the *Redstone* electronics and the launch complex, caused by the different weight of the *Mercury* capsule compared to *Atlas*'s normal payload of weapons. The weight difference caused a slightly delayed mechanical disconnection time for a shutoff signal to the

Redstone engines, resulting in the firing of the capsule escape system. MSFC engineers quickly fixed the problem, and the next test flight, in December 1960, was a success.[47]

These failures caused STG engineers to reevaluate their design philosophy. In a report to headquarters, they stated, "It has become obvious that the complexity of the capsule and the booster automatic system is compounded during the integration of the systems."[48] Engineers from the STG, the air force, and the contractors investigated the causes of the numerous failures and frantically improved the design of the boosters, the capsules, and the interfaces between them. One important way in which NASA and the air force formalized these relationships and processes was through the development of interface specifications that defined electrical and mechanical connections between the capsule and the launch vehicle, along with vibration and acceleration loads.[49]

Mercury flight failures led to increased attention to *Gemini* and *Apollo* interfaces. In June 1963, the air force, NASA, Martin, and McDonnell initiated investigations of the structural and electronic interfaces between *Titan* and the *Gemini* spacecraft. Gemini managers created the Systems Integration Office in February 1964 to monitor spacecraft weight and interfaces between the spacecraft, the launch vehicle, and the *Agena* target vehicle with regular coordination meetings known as Interface Control Panels. Interface Specification and Control Documents recorded the results of these meetings. Engineers developed new tests, including electronic-electrical interference tests, joint combined systems tests, flight configuration mode tests, and wet mock simulated launch tests, all of which verified interfaces and launch procedures on the launch pad. Apollo personnel began to work on interface problems in July 1963, when the Panel Review Board standardized Interface Control Documents and made MSFC the Apollo document repository.[50]

The STG and its successor, MSC, evolved from NACA's hands-on research culture. Despite massive growth, the STG maintained informal mechanisms for coordination and control. MSFC too held fast to its discipline-based army heritage. Both organizations benefited from military and industrial processes used by their contractors. Because of their prestige and deep pockets, the manned programs commanded attention from everyone, from the top of the management hierarchy to the shop floor workers at contractor facilities.

Both MSC and MSFC were ultraconservative concerning their own designs. They emphasized reliability and safety above cost, allowing costs to increase. The exemplary flight records of Mercury and Gemini showed that this conservatism could help ensure that technical objectives were achieved. Considering that the air force was then instituting a program to improve *Atlas*'s reliability to 75%, achieving a perfect flight record was no easy feat.[51] However, the field centers' proclivity to increase costs was a weakness soon exploited by Congress and NASA headquarters to tighten control.

The Cost Crisis of 1962–1964

In technical terms, NASA's organization and procedures proved successful on Mercury and Gemini. Both programs boasted enviable flight records, completing their objectives and successfully returning their astronauts on every flight. On the other hand, both programs featured large cost increases. McDonnell's *Mercury* capsule contract grew at an astounding rate, from an initial contract of $19,450,000 in 1959 to a total of $143,413,000 by the program's end in 1963. Within months of its start, Gemini was headed for large cost increases as well. NASA funded the increases, although Congress began to ask questions in the fall of 1962.[52]

OMSF's Holmes, overseeing the three manned space flight programs, saw the escalating cost trends as well as anyone. A hard-nosed project manager from Radio Corporation of America's (RCA's) Ballistic Missile Early Warning System, Holmes realized that something had to be done if Apollo was to meet Kennedy's 1969 deadline to land a man on the Moon. Holmes also recognized that Kennedy had staked his reputation on the manned program and would go to great lengths to ensure its success. Based on this, he felt no particular need to control costs and instead requested more funds from NASA Administrator Webb.[53]

Holmes first asked Webb to go back to Congress for a supplemental appropriation. Despite Kennedy's support, Webb recognized that Congress was becoming uneasy about NASA's burgeoning costs, and he refused. Holmes next demanded that Webb strip other NASA programs to support the manned program. When Webb again refused, Holmes went over his head, appealing directly to President Kennedy. This infuriated Webb, now placed in the uncomfortable position of justifying why he should not strip other NASA projects to fund Kennedy's priority program. Although Kennedy was not sure that he "saw eye-to-eye" with Webb on this issue, he backed his chosen administrator. Webb replaced his insubordinate OMSF director with STL executive George Mueller (pronounced "Miller").[54]

Under the circumstances, further cost pressures were unwelcome. In March 1963, Gemini project manager James Chamberlin admitted that the project required another huge infusion of money. The unwelcome news led to his replacement by Charles Mathews. Mathews immediately set up a committee to improve cost estimation at NASA and McDonnell. After another year of strained budgets and organizational crises—invariably related to novel portions of the project—Gemini's budget stabilized, leading to an enviable record of flight success. However, cost estimates had risen from $531,000,000 to $1,354,000,000 between 1962 and 1964.[55]

With Congress increasingly concerned with costs, Webb instigated an internal investigation of NASA projects in early 1964.[56] Deputy Associate Administrator Earl Hilburn chaired the investigation to study NASA's project scheduling and cost estimating methods. In confidential reports submitted in September and December of 1964, Hilburn described NASA's abysmal record on cost and schedule control. Engine development showed by far the largest schedule slips, at 4.75 times the original estimate. Hilburn found significant slips in launch vehicles requiring new engines (3.09), manned spacecraft (2.86), "simple" scientific satellites (2.85), and astronomical observatories (2.83). Classified by field center, he found that the Jet Propulsion Laboratory (JPL) had the smallest schedule slips (1.6) and MSC the largest (3.06). The newest centers, MSC and Goddard, showed the largest slips, and the older centers (JPL, Langley) showed the smallest. Costs varied similarly.[57]

Hilburn noted that the DOD experience was also "one of over-runs and schedule slippages in the majority of cases." He concluded that these results were behind recent changes in DOD procurement, "including program definition, and incentive contracting." Based on DOD precedents, NASA had already begun converting contracts from cost-plus-fixed-fee contracts to incentive contracts that awarded firms a higher fee when they performed well and a lower fee when they did not. The Hilburn Report lent support to contract conversion at NASA, which NASA implemented from 1964 through 1966.[58] NASA also adopted the DOD practice of phased planning, requiring contractors to better predict costs and schedules during a definition phase of development. The directive had little initial impact because of disagreements about implementation, and more importantly, because few new projects started in 1966 and 1967.[59]

NASA's manned space programs were among the worst offenders as NASA's budget exploded between 1962 and 1964. Having grown accustomed to generous, even extravagant budget increases without justification or congressional concern, NASA's continued cost excesses were understandable, if not justifiable. As in other organizations, the time came when its initial growth surge and justification had to slow and managers had to predict and hold to funding profiles. At the highest levels, NASA could replace recalcitrant executives and implement new standards and processes, but to truly predict and control costs, NASA would have to implement rigorous procedures on its research and development (R&D) programs.

Systems Management, Air Force Style

The informal management methods that characterized the manned program's first few years did not last. New OMSF chief George Mueller understood very well that he would have to implement rigorous cost prediction and control methods to address NASA's unruly engineers and R&D projects. To do this, he would turn to the management methods with which he was familiar, those developed in the air force and at Thompson-Ramo-Wooldridge (TRW).

Mueller began his career in 1940 working on microwave experiments at Bell Telephone Laboratories. After the war he taught electrical engineering at Ohio State University, and in 1957 he joined Ramo-Wooldridge's STL as the director of the Electronics Laboratories. He moved up quickly, becoming program manager of the Able space program, vice president of Space Systems Management, and then vice president for R&D. Mueller helped build the air force's bureaucratic system to control large missile and space projects.[60]

Before Mueller started his NASA duties, he performed his own investigation of OMSF. His first impression of NASA was that "there wasn't any management system in existence." The many interface committees and panels worked reasonably well but did not penetrate "far enough to really be an effective tool in integrating the entire vehicle." Most seriously, Mueller found no means to determine and control the hardware configuration, leaving no way to determine costs or schedules. Mueller concluded that he had to "teach people what was involved in doing program control."[61]

In August 1963 Mueller invited each of the field center directors to visit him. He explained his proposed changes and how they would help solve NASA's problems with the Bureau of the Budget and the President's Science Advisory Committee. Mueller explained, "If we didn't work together, we were sure going to be hung apart." Mueller had little trouble convincing *Apollo* spacecraft manager Joseph Shea at MSC that his changes were necessary, given Shea's prior experience at TRW. However, he met resistance from MSFC leaders. When MSFC manager Eberhard Rees challenged Mueller's proposals, Mueller retorted that "Marshall was going to have to change its whole mode of operation." Two weeks later, at another MSFC meeting, von Braun gave Mueller "one of his impassioned speeches about how you can't change the basic organization of Marshall." Refusing to back down, Mueller told him that MSFC's laboratories "were going to have to become a support to the program offices or else" they "weren't going to get there from here." Von Braun, in response, reorganized MSFC on September 1, 1963, to strengthen project organizations through the creation of the Industrial Operations branch.[62]

Webb strengthened Mueller's position by making the directors and projects at MSC, MSFC, and KSC report to OMSF. Mueller reduced attendance at the Manned Space Flight Management Council to himself and the center directors to ensure coordinated responses to OMSF problems. Borrowing from the air force Minuteman program, Mueller also formed the Apollo Executive Group, which consisted of Mueller and Apollo contractor presidents. The group members met periodically at NASA facilities, where Mueller pressured them to resolve problems.[63]

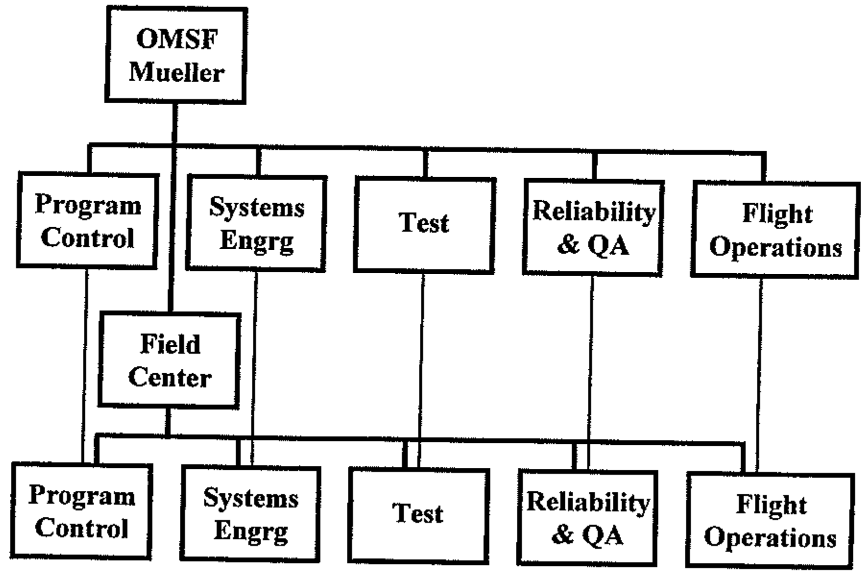
Facing the same dilemma Holmes faced—Apollo's slipping schedules and cost overruns—Mueller concluded that the only way to achieve the lunar landing before 1970 within political and budget constraints was to reduce the number of flights. In his "all-up testing" concept, each flight used the full Apollo flight configuration. This approach, used on the Titan II and Minuteman programs, violated von Braun's conservative engineering principles. Von Braun's existing plan used a live *Saturn* first stage with dummy upper stages for the first test. The second test included a live second stage with a dummy third stage, and so on. By contrast, the all-up concept used all flight stages on the very first test. This reduced the number of test flights and eliminated different vehicle configurations, with their attendant differences in designs, ground equipment, and procedures.[64]

Saturn V program manager Arthur Rudolf cornered Associate Administrator Robert Seamans at a meeting, showing him a model of the *Saturn V* dwarfing a *Minuteman* model, saying, "Now really, Bob!" Seamans got the hint—Mueller's concept did not apply to the more complex *Saturn V.* Encouraged, Rudolph showed the same models to Mueller. Mueller replied, "So what?" The all-up concept prevailed.[65]

In November 1963, Mueller reorganized the Gemini and Apollo Program Offices, creating a "five-box" structure at headquarters and the field centers. The new structure (see figure) ensured that the field centers replicated Mueller's concept of systems management and provided Mueller with better program surveillance. Inside these "GEM boxes,"[66] managers and engineers communicated directly with their functional counterparts at headquarters and other field centers, bypassing the field centers' normal chain of command. As one NASA manager put it, "Anywhere you wanted to go within the organization there was a counterpart whether you knew him or not. Whether you had ever met the man, you knew that if you called that box, he had the same kind of responsibility and you could talk to him and get communication going."[67]

Mueller's new organization initially wreaked havoc at NASA headquarters, because the change converted NASA engineers who monitored specific hardware projects into executive managers responsible for policy, administration, and finance. For several months after the change, headquarters was in turmoil as the staff learned to become executives.[68]

NASA's organizational structure changed as a result of Mueller's initiatives, but he could not always find personnel with the management skills he desired. Shortly after assuming office, Mueller wrote to Webb, stating that NASA could use military personnel trained in program control. Because of the air force's interest in a "future military role in space," Mueller believed that it would agree to place key personnel in NASA, where they would acquire experience in space program management and technology. NASA, in turn, would bene-



George Mueller's "five box" structure replicated the headquarters OMSF organization through the OMSF field centers. Adapted from "Miscellaneous Viewgraphs," circa January–February 1964, "Organization and Management," folder LC/SPP-43:11.

fit from their contractual experience and program control methods. Mueller stated, "It is particularly worth noting that the Air Force, over a period of years, has developed the capability of managing and controlling the very contractors upon whom we have placed our primary dependence for the lunar program." He proposed to place Minuteman program director Phillips in the position of program controller for OMSF and contacted AFSC chief Schriever regarding this assignment. Schriever agreed, but only on the condition that Phillips become Apollo program director. From this position, Phillips would transform NASA's organization and would become known as Apollo's Rock of Gibraltar.[69]

Phillips surmised, "NASA had developed to be a very, very professional technical organization, but they had almost no management capability nor experience in planning and managing large programs."[70] Phillips turned to the air force for reinforcements and to his most valuable tool from Minuteman, configuration management.

In a January 1964 letter to Schriever, Phillips asked for further air force per-

George Mueller (left) and Samuel Phillips (right) imposed air force management methods on Apollo by introducing new procedures and bringing dozens of air force officers into NASA's manned space flight programs. Courtesy NASA.

sonnel to man OMSF's program control positions. After AFSC assigned two officers to NASA, Phillips created a list of fifty-five positions that he wanted to fill with air force officers. Such a large request entailed formal negotiations between NASA and the air force for Phillips's "Project 55." Secretary of the Air Force Eugene Zuckert agreed to consider transfers if NASA better defined the position requirements, so that he could ensure that the positions would enhance the officers' careers. In September 1964, the joint NASA–air force committee reported that ninety-four air force officers already worked in NASA and that forty-two of the fifty-five additional positions requested should be filled by further air force assignments. The air force reserved the right to select junior and midgrade officers for NASA tours of at least three years. Eventually, Phillips requested and received assignments for 128 more junior officers, who were mainly assigned to Apollo operations in Houston.[71]

Mueller and Phillips placed officers in key managerial positions throughout the Apollo and Gemini programs, particularly in project control and configuration management offices. Phillips also requested a few senior officers by name. By December 1964, NASA and the air force agreed to assign Phillips as Apollo program director. NASA assigned Brig. Gen. David Jones as deputy associate administrator for Manned Space Flight (Programs), Col. Edmund O'Connor as director of MSFC Industrial Operations, Col. Samuel Yarchin as deputy director of the Saturn V Project Office, and Col. Carroll Bolender as Apollo mission director.[72]

Phillips recognized that NASA's engineers would resist his primary control technique, configuration management. To meet this issue head-on, he scheduled a Configuration Management Workshop in February 1964 in Los Angeles. Each NASA and contractor organization could invite two or three people, except MSFC, which could invite six. At the conference, Phillips stated, "Coming out of Minuteman and into Apollo, I think I've been identified as a procedures and methods man with a management manual all written that I intend to force on the Apollo program and down the throats of the existing 'good people' base. So I ask myself, 'WHY PROCEDURES AND METHODS?'" Phillips noted that good processes and methods made good people's work even better and that they enabled the manager to communicate and control more effectively because they created a "high percentage of automatic action." Because of Apollo's rapid growth, the program needed better communications between NASA and contractor organizations, and all parties had to use consistent language. Phillips stated that good procedures had a high probability of preventing oversights or shortcuts that could lead to catastrophe. He noted, "The outside world is critical, including the contractors, Congress and the GAO [General Accounting Office], and the press."[73] Solid procedures would help to protect NASA from criticism, by preventing failures and by documenting problems as they were found.

Phillips explained his system of design reviews and change control, both of which would help managers control resources. He viewed the field centers as Apollo prime contractors and considered configuration management a contractual mechanism to control industry. Phillips proposed strong project management along with a series of reviews tied to configuration control. Consistent with military doctrine, Phillips believed that "diffused authority and responsibility" meant a "lack of program control," so he assigned responsibility for each task to a single individual and gave that person authority to accomplish the task.[74]

To aid his initiative, Phillips modified Apollo's information system. In the existing headquarters program control and information system, data were collected from the field centers each month, and then headquarters managers reviewed the data for problem areas and inconsistencies, at which time they advised the centers of any problems. Instead, Phillips wanted daily analysis of program schedules and quick data exchange to resolve problems, with data placed in a central control room modeled on Minuteman. He immediately placed contracts for a central control room, which eventually contained data links with automated displays to Apollo field centers.[75]

At MSFC, Saturn managers already had a control room to track official program activities. Saturn V manager Arthur Rudolf assigned names to each chart on the control room wall, so that whenever he found a problem, he could immediately call someone who understood the chart's details and implications. The charts varied over time, as new ones representing current problems and status appeared, replacing charts addressing problems that had been resolved.

This room, although useful, had its problems. Rudolf did not like PERT and reverted instead to "waterfall" charts (Gantt, or "bar," charts arranged in a waterfall fashion over time). Project control personnel translated PERT charts into Rudolf's waterfall charts, located in a small room across the hall. Because the control room displayed the "official" status of items, it did not always have the kind of information Rudolf wanted. A typical problem would be that a hardware item might be completed but the paperwork might still be incomplete. The control room information would not be updated until the paperwork was complete, and hence it did not reflect the true status of the hardware. By contrast, the "mini control room" across the hall had "grease-penciled" charts with more up-to-date information. With MSFC personnel calling contractors to get status updates, this mini control room buzzed with activity.[76]

Phillips devoted great effort to the promotion of configuration management. His headquarters group worked with the air force to develop the *Apollo Configuration Management Manual.* Issued in May 1964, this manual copied the air force's manual, which AFSC officers were updating at the time. He

wrote letters to the field center directors emphasizing the manual's importance and directed them to develop implementation plans. By fall 1964, the field centers were actively creating configuration control boards (CCBs), developing the forms and procedures, and directing contractors to implement configuration management.[77]

Configuration management required that NASA have firm requirements and specifications for *Apollo,* which did not yet exist, despite three-year-old contracts between NASA and its contractors. Phillips ordered NASA headquarters and systems contractor Bellcomm to develop definitive specifications. Not wanting Bellcomm to take over this task, field center and contractor personnel quickly became involved, resulting in firm specifications for all Apollo contracts.[78]

NAA recognized that with Phillips as Apollo program manager, failure to enhance configuration management "could well be a serious mistake," so it led a study of the process. Under Phillips's plan, NASA placed preliminary specifications under change control after the Preliminary Design Review, and the hardware under change control after the First Article Configuration Inspection. NAA's group discovered that after the Critical Design Review, NASA did not elevate the final design specifications to contractual status, which could lead to contractual disagreements over design specifications changes. NAA raised this to NASA's attention, resulting in further enhancement of configuration management.[79]

Phillips soon encountered the resistance he expected. Some of it was passive. He tried to educate NASA personnel about systems management through air force project management and systems engineering courses at the Air Force Institute of Technology, where he occasionally lectured. Despite his enthusiasm, only a handful of NASA engineers and managers attended the one-week course.[80]

Other resistance was overt. At the June 1964 Apollo Executives Meeting, Phillips had his headquarters configuration control manager describe configuration management to the field center directors and industry executives. After the presentation, the NASA field center directors argued against Phillips's plan.[81] MSC Director Gilruth had heard that configuration management cost one additional person for every manufacturing team member. If this was true, then configuration management would be far too expensive.

Phillips's team replied that based on Minuteman experience, configuration control required only one person per one hundred manufacturing team members. Air force configuration managers described how Douglas Aircraft took four months to confirm the *S-IV* stage configuration prior to testing and delivery to NASA, whereas on the Gemini Titan II, a program with configuration management, it took two days.[82]

MSFC Director von Braun objected: "This whole thing has a tendency of moving the real decisions up, even from the contractor structure viewpoint, from the one guy who sits on the line to someone else." Boeing President Bill Allen replied, "That's a fundamental of good management." After all, he said, "Who, around this table, makes important decisions without getting advice from the fellow who knows?" Von Braun retorted, "The more you take this into the stratosphere and take the decisions away from the working table—I think the object of this whole thing is to remove it from the drawing board." Allen, whose company had originally created configuration management, replied that you merely had to move the "top engineering guy into the position of the Configuration Manager."[83]

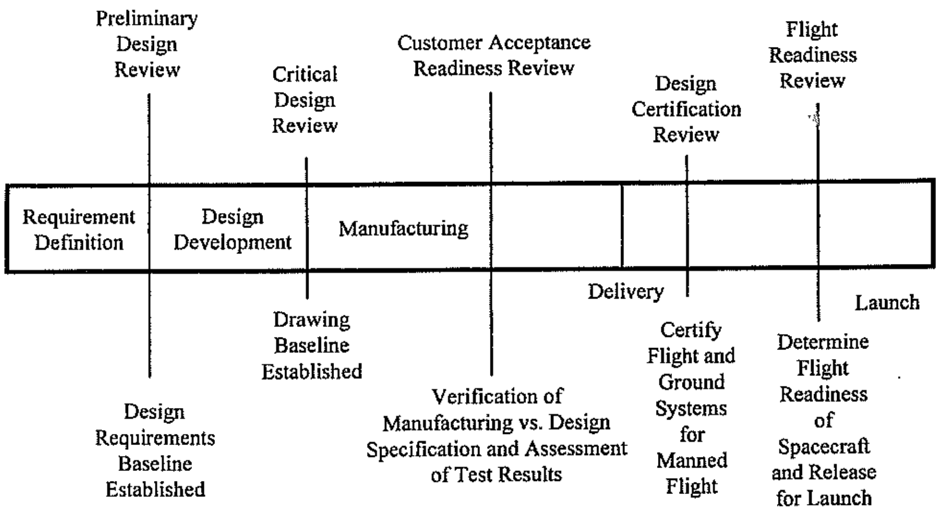
Frustrated in this argument, von Braun retorted that because the military produced a thousand Minutemen, whereas there was only one or just a few Apollos, "We have to retain a little more flexibility." Again, Boeing's Allen disagreed: "Maybe I don't understand, but in my simple mind, it doesn't make any difference with respect to what has been outlined here, whether it's R&D, *Saturn,* or whether you're trying to produce a thousand letters." Von Braun replied, "If you want to roll with the punches, then you have to maintain a certain flexibility." OMSF chief Mueller intervened; on the Titan III program, the first with configuration management from the start, Mueller noted, "Everything is on cost and schedule, even though it married solids and liquids."[84] Mueller concluded, "Configuration management—doesn't mean you can't change it. It doesn't mean you have to define the final configuration in the first instance before you know that the end item is going to work. That isn't what it means. It means you define at each stage of the game what you think the design is going to be within your present ability. The difference is after you describe it, you let everybody know what it is when you change it. That's about all this thing is trying to do."[85]

Mueller, who had the ultimate authority to force implementation, quelled executive objections, but resistance continued in other forms. MSC assigned only one person to configuration management by October 1964, slowing its adoption there and at MSC's contractors. There was continued engineering resistance to change control, Phillips noted: "Engineers always know how to do it better once they've done it, and want to make their product better." Yet "even engineers will admit that changes first of all must be justified," he added.[86]

Contractors had one last chance to charge additional costs to the government before NASA could control them in detail using configuration management. They used the opportunity, preventing full implementation into late 1965 by charging high rates to implement configuration control systems. NASA auditors found numerous deficiencies, including incomplete engineering release systems, no configuration management of major subcontractors, uncontrolled test requirements and procedures, poor numbering systems, lack of documentation, and, in a few cases, no system at all.[87]

With continued exhortation and substantial pressure, Phillips established configuration management on Apollo by the end of 1966, with four different levels of change control and authority: contractor, stage and system manager, program manager, and program office. Contractors could authorize changes that did not affect any other contractors or specifications. Stage and system managers could authorize changes within their own systems. Only program manager offices at the field centers could authorize changes affecting interfaces between stages, systems, or field centers. Phillips in the Program Office authorized changes to the master schedule, hardware quantities, or the overall program specifications. The full system included six formal reviews, after which NASA approved or modified the specifications, designs, and hardware.[88]

Because the Apollo program had been under way for three years before Phillips took control, many of the designs never underwent the initial reviews called for in the new system. NASA had awarded Apollo contracts without accurate specifications, revised the design after contract awards, and designed and even tested system components without specifications. Only one element, the Block II command module for the lunar orbit and landing missions, fol-



Phillips's review processes for Apollo, which he used to ensure the success of the moon landings. Adapted from Robert C. Seamans Jr. and Frederick I. Ordway, "The Apollo Tradition: An Object Lesson for the Management of Large-Scale Technological Endeavors," in Frank P. Davidson and C. Lawrence Meador, eds., *Macro-Engineering and the Future: A Management Perspective* (Boulder, Colo.: Westview Press, 1982), 20.

lowed Phillips's process in its entirety.[89] Other vehicle elements began their first design reviews at their state of maturity when Phillips levied his requirements.

Phillips augmented configuration control with other information sources. He held daily and monthly meetings with program personnel. In turn, Mueller and Seamans reviewed Apollo each month, and Webb reviewed it annually. The project developed a computer system to automate failure reports, cost data, and parts information. Apollo's Reliability and Quality Assurance organization developed into an important management tool, supplying information on reliability, test results, and part defects as well as current plans, status reports, schedules, funding, and manpower. It forwarded quarterly and weekly highlight reports on each major system element.[90]

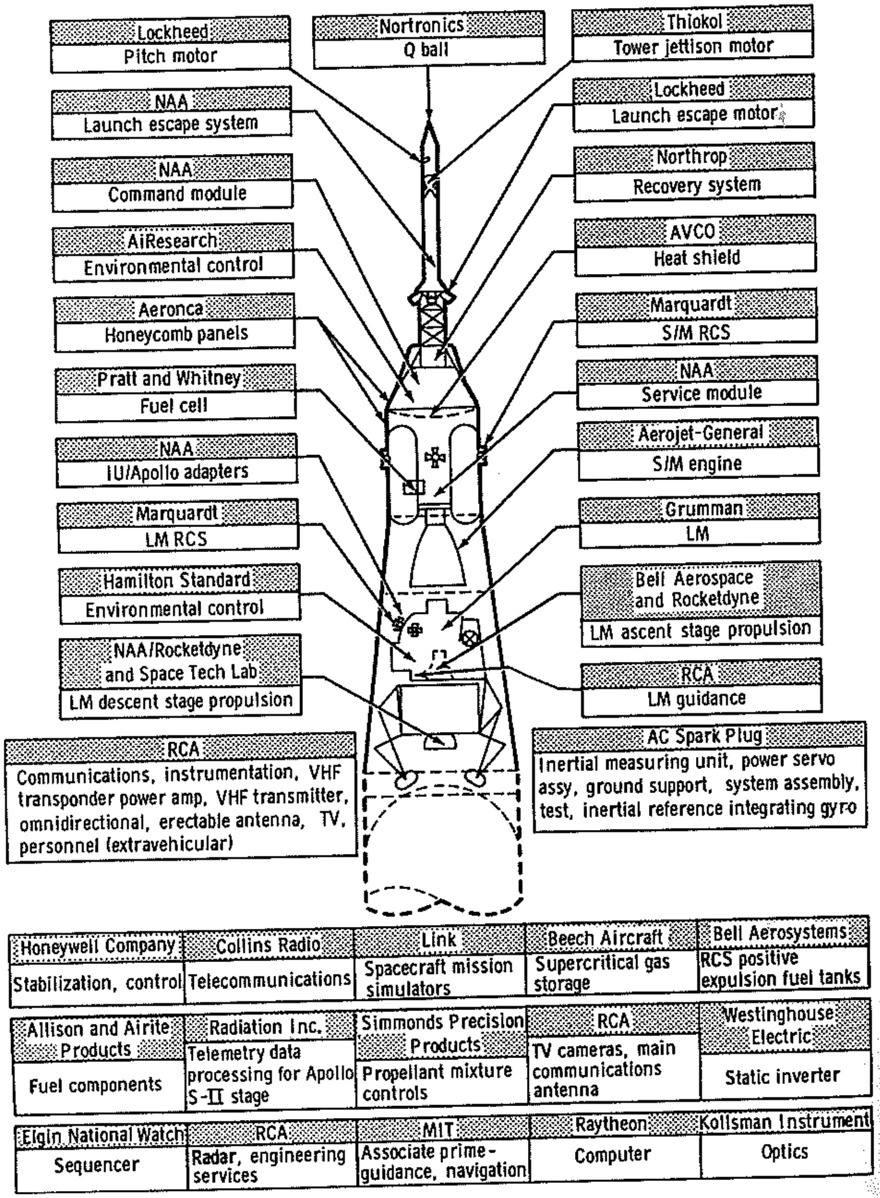
The two years following the hiring of George Mueller in September 1963 marked Apollo's transition from a loosely organized research team to a tightly run development organization. Mueller made important early decisions, including instituting his GEM box organization formalizing systems engineering, reliability and quality assurance, and project control on the manned programs. Mueller forced von Braun and Gilruth to adhere to his all-up decision, sharply reduced flight tests in favor of ground testing, and gave more responsibility to contractors. Finally, he started the importation of air force officers to implement program control, beginning with Minuteman director Phillips.

Phillips brought in air force officers to implement configuration management. Configuration management required precise knowledge of the system specifications and design, the baseline against which managers and systems engineers judged changes. This, in turn, required a series of design reviews and managerial checkpoints that progressively elevated specifications and designs to controlled status. Despite resistance, by 1966 Mueller and Phillips augmented NASA's processes by firmly establishing air force methods within OMSF. The new management methods could not prevent all technical problems or make up completely for the earlier lack of management control, nor did contractors uniformly enforce them. For most technical programs, the most difficult times involve testing problems that arise. Apollo would be no different.

Smoke, Fire, and Recovery

Apollo's troubles began in September 1965, when NAA's second stage ruptured during a structural test.[91] Engineers pinpointed the fault, and in the process MSFC managers concluded that NAA's management was to blame for shoddy workmanship. By October, the Industrial Operations manager, Brig. Gen. Edmund O'Connor, told von Braun, "The S-II program is out of control." He believed its management was to blame. O'Connor was equally blunt in a letter to Space and Information Systems Division (S&ID) President Harrison Storms: "The continued inability or failure of S&ID to project with any reasonable accuracy their resource requirements, their inability to identify in a timely manner impending problems, and their inability to assess and relate resource requirements and problem areas to schedule impact, can lead me to only one conclusion, that S&ID management does not have control of the Saturn S-II program."[92]

Phillips went immediately to NAA with a "tiger team" of nearly one hundred NASA personnel to "terrorize the contractor,"[93] reporting the team's



Apollo with its major contractors identified. Apollo was perhaps the largest single R&D project of all time, integrating many contractors for its stages and requiring massive launch and operations facilities and organizations. Saturn V contractors not identified. Courtesy NASA.

findings in December 1965 in what later became known as the Phillips Report. While writing to NAA that "the right actions now" could improve the program, Phillips privately wrote Mueller that NAA's president was too passive. Storms, Phillips said, should "be removed as president of S&ID and be replaced by a man who will be able to quickly provide effective and unquestionable leadership for the organization to bring the division out of trouble."[94]

NAA responded by placing Gen. Robert Greer, retired from the air force, in charge of the S-II program. Greer updated the management control center and ensured more rapid exchange and collection of information through Black Saturday meetings modeled after those in Bernard Schriever's ballistic missile program. Greer also instituted forty-five-minute meetings every morning, eventually cutting back to twice a week. Greer's reforms began to take hold but did not prevent the May 1966 loss of another test stage because of faulty procedures. NASA clamped down further, requiring NAA to develop better methods for managing and planning its work. In the summer of 1966, after two years of studies and preparation, NAA deployed work package management for the S-II and Command and Service Module.[95]

Work package management extended project management to lower project levels and combined accounting and contracting procedures by creating a specific work package for each program task. The company assigned responsibility for each task to one person, a mini project manager for the task who accounted for performance, cost, and schedule in the same way and with the same tools as the overall project manager. Each work package was a "fundamental building stone," with specifications, plans, costs, and schedules to help managers in their monitoring. Prior to the development of work packages, "It was difficult to say what manager was responsible for a particular cost increase because there were 10 or 15 functional and subcontractor areas involved."[96] In later versions, the work package numbering scheme matched that for cost accounting.

Grumman's difficulties on the lunar module also attracted NASA attention. Troubles first appeared in schedule slips on its ground support equipment in the spring of 1966. Alarmed at Grumman's growing costs, Phillips sent a management review team to Grumman that summer, prompting Grumman to sack the program manager, establish a program control office, and move Grumman's vice president to the factory floor to monitor work. By fall, NASA pushed Grumman into adopting work package management.[97] It did not immediately solve Grumman's difficulties. The primary problem was a late start due to NASA's delayed decision to use lunar orbit rendezvous. However, work package management and the new program control office found and resolved problems more quickly than before.

Despite these difficulties, Apollo moved briskly forward until its most severe crisis struck on January 27, 1967. That day, astronauts and KSC personnel were performing tests in preparation for launch of the first manned Apollo mission. At 6:30 that evening, the three astronauts scheduled for that mission, Virgil Grissom, Edward White, and Roger Chaffee, were in the spacecraft command module testing procedures. At 6:31, launch operators heard a cry from the astronauts over the radio, "There is a fire in here!" Those were their last words. All three astronauts died of asphyxiation before launch personnel reached them.[98]

KSC personnel immediately notified NASA headquarters. Administrator Webb hurriedly planned for the political fallout. He sent Seamans and Phillips to Florida, while he persuaded the president and Congress to let NASA perform the investigation.[99] NASA's investigation concluded that the causes of the disaster were faulty wiring, a drastic underestimation of the dangers of an all-oxygen atmosphere, and a capsule design that precluded rapid escape. No one had realized how dangerous the combination was. NASA had used a pure oxygen atmosphere in all of its prior flights, as did air force pilots in their high-altitude flying. As Col. Frank Borman, one of NASA's most experienced astronauts, put it during the Senate investigation, "Sir, I am certain that I can say now the spacecraft was extremely unsafe. I believe what the message I meant to imply was that at the time all the people associated and responsible for testing, flying, building, and piloting the spacecraft truly believed it was safe to undergo the test."[100]

Congress did not prove NASA to be negligent or incompetent. One of the investigation's important results was a nonfinding. Despite searching long and hard, Congress did not find fault with Phillips's management system. Phillips had already uncovered problems with NAA and had been working for some time to make improvements to its organization and performance. The management system used to organize the capsule design was NASA's original committee-based structure, upon which Phillips had superimposed configuration management. He and his management system came out unscathed.

Congressional investigations did uncover some of NASA's dirty laundry, particularly problems with command module contractor NAA. Sen. Walter Mondale of Minnesota learned of the Stage II Phillips Report and confronted Webb about problems between NASA and NAA. Caught by surprise, Webb said he did not know of any such report, which at that moment he did not. After the hearing, he found out about it from Mueller and Phillips. Furious, Webb launched a "paper sweep" to search for more skeletons in the closet. The sweep uncovered a memo written by GE to Apollo spacecraft director Joseph Shea, warning Shea of the danger of fire in the command module. Shea had passed the memo on to his safety and quality assurance people, who responded that no significant dangers existed. GE, already in a sensitive situation because MSC considered it to be spying for headquarters, did not push it any further.[101]

Webb reacted angrily to these revelations. He believed OMSF had been far too independent and secretive. Webb told Seamans, "You have to penetrate the [OMSF] system, don't let Mueller get away with bullshit." The problem, according to Webb, was a lack of supervision by NASA's executive management. Mueller had "followed the policy in Houston of obtaining the very best men they could for the senior positions, and had, as a part of the process of obtaining them, given assurances that they would have almost complete freedom in carrying out their responsibilities."[102]

After Seamans left NASA in late 1967, Webb expressed shock at the poor management system.[103] Webb probably did not realize how decentralized NASA's management really was. Executive managers routinely delegated most decisions to lower levels. In the wake of the fire, this did not seem wise.

When NAA refused to make swift and comprehensive changes—and even expected to be paid a fee for the burned-out spacecraft—Webb called Boeing to see if it would take the job. Boeing said that although it did not want to take over the job, if pressed it would do so. Webb returned to NAA, demanding that it remove S&ID head Storms, further centralize Apollo project management, eliminate any fee for the failed spacecraft, and pay for improvements. NAA did not take the chance that he was bluffing. NAA was extremely unhappy with the entire situation because from its viewpoint, NASA was at fault. Shortly after contract award, over NAA's objections, NASA had directed a change from a nitrogen-oxygen atmosphere to an all-oxygen atmosphere.[104]

One problem uncovered during the investigation was GE's unwillingness to contest NASA over safety issues with a pure oxygen atmosphere. At the heart of the problem was industry's reluctance to confront NASA when industry was dependent on government funding. Despite his substantial political acumen, Webb appeared not to comprehend this. He had hired GE and Bellcomm to strengthen headquarters' ability to monitor the field centers in 1962; after the fire, Webb repeated his mistake by expanding Boeing's role from integrator of the Saturn V to integrator of the entire Apollo-Saturn system to "penetrate the OMSF system." Phillips, who understood the political problems inherent in the GE and Boeing integration efforts, revised the Boeing contract to avoid the negative consequences of Webb's misconception.[105] In essence, Webb wanted to use GE, Bellcomm, and Boeing as an arm of NASA headquarters to control MSC, MSFC, and KSC. This could not work because these contractors could not challenge NASA field center personnel for fear of losing their contracts.

Boeing, as part of its contract, further integrated the management system. The "teleservices network" connected NASA project control rooms with hard copy data transmittal, computer data transmission, and the capability to hold a teleconference involving MSC, MSFC, KSC, Michoud (where the *Saturn I* was manufactured), and Boeing's facility near Seattle. Boeing copied MSFC's program control center design at each facility.[106]

After the fire, NASA placed even more emphasis on achieving high quality and safety through procedural means. In September 1967, NASA set up safety offices at each field center, along with the first project safety plan. The next month, MSC established a Spacecraft Incident Investigation and Reporting Panel to look into anomalies. A month later, NAA created a Problem Assessment Room to report and track problems.

Phillips ordered an astounding array of program reviews to prepare for Apollo's upcoming missions. He wrote to field center managers to ensure that they used the upcoming Design Certification Reviews to evaluate all potential single-point failures.[107] In January 1968, he ordered a complete system safety

review, analyzing the interaction of the mission with the hardware, astronauts, ground systems, and personnel. Other reviews included those for quality and metrology, launch vehicle and spacecraft schedules, the communications network, flight readiness, mission planning, subcontractors, site selection, the Lunar Receiving Laboratory, flight evaluations, anomalies, crew safety, interface management, software, and lunar surface activities.[108]

NAA's procedures exemplified the upgraded problem reporting system. Engineers reported failures on a Problem Action Record form. Reliability engineers sent failed components to the appropriate organization, which responded by filling out a Failure Analysis Report describing the physical cause of the failure and the corrective actions taken or recommended. If the organization determined that an engineering change was necessary, it submitted a change request to the change boards. The program control center tracked report status, and a centralized reliability "data bank" recorded the problem and its resolution. Follow-up failure reports and dispositions closed all failure reports.[109]

Another change in the aftermath of the fire was a further strengthening of configuration management, primarily through changing CCB operating procedures. An October 1967 rule disallowed nonmandatory changes for the first command and lunar modules and required the MSC Senior Board to rule on any and all changes to these spacecraft. A February 1968 ruling required managers to consider software changes and their ramifications in CCBs. In May 1968, Apollo Spacecraft Manager George Low specified that the MSC CCB had authority over all design and manufacturing processes.[110]

By 1968, tough CCB rules slowed the program as trivial changes came to the attention of top managers. Eventually, even Phillips realized that centralization through configuration management could go too far. In September, MSC managers classified changes into two categories: Class I changes, which MSC would pass judgment upon, and Class II changes, which could be approved by the contractors. Classification did not by itself help much, so in October 1968 Phillips gave Level II CCBs more authority, while higher levels ruled on schedule changes.[111]

The Apollo project met its technical and schedule objectives, landing men on the Moon in July 1969 and returning them safely to Earth. Anchored by configuration management, Phillips's system weathered the storm of problems uncovered through testing and Apollo's most severe crisis, the 1967 death of the three astronauts and the ensuing investigations. Despite strenuous efforts, congressional critics did not find many flaws with Phillips's management scheme and concurred with NASA that the fire resulted from a tragic underestimation of the danger.

Configuration management was Phillips's most powerful tool. Whenever problems occurred, his almost invariable response was to strengthen configuration management. Having found that his favorite method could be overused, by the end of 1968, Phillips gave lower-level CCBs more authority. Configuration management formed the heart of Apollo's system and has remained at the core of NASA's organization ever since.

Von Braun's Conversion

Despite the imposition of systems engineering and configuration management on MSFC, they remained foreign concepts to members of Huntsville's engineering family. They had been working together on rockets since the 1930s, and the many years of experience had taught them the technologies, processes, and interactions necessary to build rockets. These engineers understood rocketry and each other so well as to make formal coordination mechanisms such as systems engineering redundant. The efforts of Mueller and Phillips had brought configuration management and air force methods to contractors, but the functional, discipline-based laboratories remained the centers of power in MSFC.

As MSFC's effort on the Apollo program peaked in 1966 and layoffs threatened, however, MSFC leaders realized that they would have to diversify beyond rocketry to keep themselves in business.[112] In new fields such as manned space stations and robotic spacecraft, MSFC's unmatched ability in rockets meant little, and they soon found new utility in systems engineering.

By the summer of 1968, von Braun recognized that he needed to strengthen systems engineering at MSFC. He called in Philip Tompkins, a communications expert from Wayne State University, to study MSFC's organization and recommend how better to implement systems engineering. At the time, Mueller was pressing von Braun to emphasize systems engineering in the design of the *Skylab* space station. Von Braun explained to Tompkins that Mueller, who had been trained in electrical engineering, thought more naturally in terms of a "nervous system" than he, who thought of rockets as machines. Von Braun belatedly saw the validity of Mueller's point of view and was determined to reorient MSFC along systems engineering lines so as to better coordinate MSFC's design efforts.[113]

Tompkins investigated MSFC's organization and soon concluded that the design laboratories were overly oriented toward "low-level subsystems engineering." As one manager stated it, "If we had a lawnmower capability at the Marshall Center, we'd put lawnmowers on all the vehicles."[114] To combat this, Tompkins recommended significant strengthening of the systems engineering office. With this change, systems engineering by late 1968 became a much stronger element within MSFC, albeit weaker in the traditional rocket groups than the newer organizations that focused on other projects. As MSFC's "family" organization and expertise in rocketry grew less important, systems engineering took their place. Formal coordination processes replaced the informal methods that sufficed in von Braun's heyday.

Why did NASA's most experienced group of engineers take so long to embrace systems engineering? Three factors contributed: the almost exclusive use of in-house capability for rocket development and testing, the extraordinary continuity of von Braun's team, and the continuity of the team's R&D project. From the mid-1930s until the early 1960s, von Braun's team members relied upon their own capabilities to design rockets, using external contractors sparingly. When they did use contractors, they did so for only specific components, or they closely monitored the contractors, such as Chrysler for the Redstone and Jupiter. The use of contractors significantly increased for the Saturn V project, and systems engineering began to make inroads into MSFC at this time. However, not until MSFC diversified out of rocketry and many of the original team began to retire in the late 1960s did systems engineering become a major element of Marshall's R&D process.

The continuity of von Braun's team, along with the continuity of the technologies upon which the team worked, helps to explain the dismissal of systems engineering at MSFC. Simply put, when each team member knew the job through decades of experience and knew every other team member over that period, formal methods to communicate or coordinate were redundant. Rocket team members knew their jobs, and each other, intimately. They understood what information their colleagues needed, and when. When they began to work on new products such as space stations and spacecraft in the late 1960s, it was no longer obvious how each team member should communicate with everyone else. Formal task planning, coordination, and communication became a necessity, and systems engineers performed these new tasks.

Conclusion

Systems management evolved as the manned space programs developed. Like the ballistic missile programs before them, the manned programs were inaugurated with few cost constraints and substantial external pressure to speed development. Despite massive cost overruns, the programs continued for the first few years with few questions from headquarters or Congress. Glennan, and later Webb, let the STG, MSC, and MSFC do their jobs with minimal supervision. These organizations used informal engineering committees to manage the manned programs. When NASA needed rigor in manufacturing and component quality, it had the air force and its industrial contractors to supply them. Informal methods frequently produced technical success but failed miserably at predicting costs.

Spiraling costs led Holmes, the first head of OMSF, to challenge Webb. Holmes's failure made it obvious to his replacement, Mueller, that he had to control costs. To do so he enlisted the help of air force officers, led by Phillips. Mueller forced MSC and MSFC to adopt stronger project management, institute systems engineering, expand ground testing, and report more thoroughly to headquarters. Phillips instituted configuration management and project reviews throughout Apollo to control technical, financial, and contractual aspects as well as the scheduling of the program. Air force officers brought in by Mueller and Phillips propagated the reforms and transformed OMSF's organizations into project-oriented hierarchical development organizations.

Systems management made development costs more predictable and created technically reliable product, but at a price. The disadvantages of systems

management would become apparent later, but for the moment it was a managerial icon. If there was a secret to Apollo, it was Phillips's organizational reforms, which transferred air force methods to NASA, superimposed upon the technical excellence of STG and MSFC engineers. Europeans would eventually make a concerted effort to learn the managerial secrets of Apollo, but not before trying their own ideas, and failing miserably.

from Dr. Newell to Dr. Pickering, 13 July 1964, JPLA 6-23e. Letter from Dr. Newell to Dr. Pickering, 14 July 1964, JPLA 6-23f. Letter from Dr. Pickering to Mr. Cortright, 14 July 1964, JPLA 6-23h. Letter from Mr. Cortright to Dr. Pickering, 4 August 1964, JPLA 6-23I. Quotation from Hall, *Systems Engineering*, 8.

76. *Project Surveyor*, 26–28.

77. *Project Surveyor*, 30–35.

78. Planetary spacecraft can be launched only when the positions of Earth and the target planet are in favorable locations—in the case of Mars, about every 18 months.

79. *From Project Inception*, 14, 25, 26.

80. Ibid., 28, 30, 31.

81. Ibid., 31–35.

82. System test is the group of final tests performed between the time that the spacecraft is first fully assembled and the time that the spacecraft is launched.

83. *From Project Inception*, 35–36.

84. Koppes, *JPL*, 169–72.

85. Ibid, 181–84. Hall, *Lunar Impact*, 289–306.

86. See the next chapter for the origins of work package management at North American.

87. JPL, *Surveyor Final Project Report*, part I, Project Description and Performance, Technical Report 32-1265 (Pasadena, Calif.: JPL, 1 July 1969), JPLA 6-256, 45–46, 57–58. Quotations from D. S. Liberman, F. A. Paul, and E. F. Grant, *Failure Reporting and Management Techniques in the Surveyor Program*, SP-6504 (Washington, D.C.: NASA, 1967), JPLA 6-67, 1–16.

88. *Mariner Venus 67, Project Policy and Requirements*, EPD-350, 18 May 1966, JPL, Caltech, JPLA 8-41, II-8, -9, III-1, IV-1–IV-9.

89. *Mariner Venus 67, Project Policy and Requirements*, III-1–III-5, III-17–III-20; quotation at III-3.

90. *Mariner Venus 67, Project Policy and Requirements*, Sections V, VI. Koppes, *JPL*, 196.

91. John Small, Deputy Chief, Systems Division, "Systems Engineering in Space Exploration," in *Seminar Proceedings, Systems Engineering in Space Exploration, May 1-June 5, 1963*, 1 June 1965, JPLA Historian Index No. 02 00221 XF, 1, 13–14.

92. W. Downhower, Chief Systems Design Section, "Systems Design"; C. R. Gates, Chief, Systems Division, "Systems Analysis"; Marshall Johnson, Chief, Space Flight Operations Section, "Space Flight Operations"; and John Small, Deputy Chief, Systems Division, "Program Engineering and Project Problems," in *Seminar Proceedings*, 15–32, 33–38, 39–44, 45–53.

Five | Organizing the Manned Space Program

T. Alexander, "The Unexpected Payoff of Project Apollo," *Fortune* 79, no. 7 (1969): 114.

1. The term "manned" space is anachronistic; women now frequently fly on space missions. However, it was the term used at the time, and no women flew on American programs of the 1960s and 1970s.

2. Alex Roland, *Model Research, The National Advisory Committee for Aeronautics, 1915–1958*, SP-4103 (Washington, D.C.: NASA, 1985). Also see NASA histories of the NACA and NASA field centers.

3. Andrew J. Dunar and Stephen P. Waring, *Power to Explore: A History of Marshall Space Flight Center 1960–1990*, SP-4313 (Washington, D.C.: NASA, 1999).

4. Henry C. Dethloff, *Suddenly, Tomorrow Came: A History of the Johnson Space Center*, SP-4307 (Washington, D.C.: NASA, 1993), 35–51. Initially, the STG was to be moved to Goddard Space Flight Center, but its size and goals led to the creation of an entirely new center in Houston.

5. Interview of Robert R. Gilruth by Michael D. Keller, 26 June 1967, "Gilruth, Robert R. Bio" folder 000782, LEK 1/6/2, NASAHO.

6. Shirley Thomas, "Robert R. Gilruth," in *Men of Space: Profiles of the Leaders in Space Research, Development, and Exploration* (Philadelphia: Chilton Company, 1962), 38–74. "Gilruth, Robert R. Bio," 63.

7. Robert R. Gilruth, "From Wallops Island to Project Mercury, 1945–1958: A Memoir," in R. Cargill Hall, ed., *History of Rocketry and Astronautics*, AAS History Series, vol. 7, part II (San Diego: AAS Publications Office, 1986), 445–76. Announcement, Key NASA Personnel Change, Director, Key Personnel Development, Director Manned Spacecraft Center, 18 January 1972, "Gilruth, Robert R. Bio" folder 000782, LEK 1/6/2, NASAHO.

8. Arnold S. Levine, *Managing NASA in the Apollo Era*, SP-4102 (Washington, D.C.: NASA, 1982), 61–64. Loyd S. Swenson Jr., James M. Grimwood, and Charles C. Alexander, *This New Ocean: A History of Project Mercury*, SP-4201 (Washington, D.C.: NASA, 1966), 120–22, 137.

9. *Project Mercury Status Report No. 2 for Period Ending April 30, 1959*, NASA, Langley STG, NASAHO, 1. *Project Mercury Quarterly Status Report No. 6 for Period Ending April 30, 1960*, NASA, STG, NASAHO, 33–34.

10. John M. Logsdon, Moderator, *Managing the Moon Program: Lessons Learned from Project Apollo*, Proceedings of an Oral History Workshop Conducted July 21, 1989, Monograph in Aerospace History No. 14 (Washington, D.C.: NASA, July 1999), 18.

11. *Mercury Project Summary, Including Results of the Fourth Manned Orbital Flight*, SP-45 (Washington, D.C.: NASA, 1963), 18–21.

12. Logsdon, *Managing the Moon Program*, 32–33.

13. *Mercury Project Summary*, 18–21.

14. Marlowe Cassetti, interview by Carol Butler, *Quest* 7, no. 1 (1999): 46.

15. Swenson et al., *This New Ocean*, 149, and *Project Mercury Status Report*.

16. Swenson et al., *This New Ocean*, 146–51, 153, 251.

17. Swenson et al., *This New Ocean*, 77–82. Courtney G. Brooks, James M. Grimwood, and Loyd S. Swenson Jr., *Chariots for Apollo: A History of Manned Lunar Spacecraft*, SP-4205 (Washington, D.C.: NASA, 1979), 4, 7, 8, 11, 12.

18. Roger Bilstein, *Stages to Saturn: A Technological History of the Apollo/Saturn aunch Vehicle*, SP-4206 (Washington, D.C.: NASA, 1980), 26–31, 79–81.

19. Brooks et al., *Chariots for Apollo*, 14–17. Ivan D. Ertel and Mary Louise Morse, *'he Apollo Spacecraft: A Chronology*, vol. 1 (through 7 November 1962), SP-4009 Washington, D.C.: NASA, 1969), 47–48, 54, 57–60.

20. John M. Logsdon, *The Decision to Go to the Moon: Project Apollo and the Vational Interest* (Cambridge: MIT Press, 1970). Stephen J. Garber, "Multiple Means :o an End: A Reexamination of President Kennedy's Decision to Go to the Moon," *Quest* 7, no. 2 (1999): 5–17.

21. Ertel and Morse, *Apollo Spacecraft*, vol. 1, 54, 101–4, 120–27. Brooks et al., *Chariots for Apollo*, 35, 38–41.

22. Bilstein, *Stages to Saturn*, 81–82, 140–41. The *S-I* was a test stage, not the same as the *S-IB*, developed for the *Saturn V* lunar vehicle by Boeing. North American Rocketdyne was North American's engine group that was also working on the *F-1* engine.

23. Bilstein, *Stages to Saturn*, 267. Dunar and Waring, *Power to Explore*, 47–48.

24. Dunar and Waring, *Power to Explore*, 153.

25. Bilstein, *Stages to Saturn*, 262–63. Phillip K. Tompkins, *Organizational Communication Imperatives: Lessons of the Space Program* (Los Angeles: Roxbury, 1992), chaps. 3–5. Dunar and Waring, *Power to Explore*, 47.

26. STG, *Project Development Plan for Rendezvous Development Utilizing the Mark II Two-Man Spacecraft*, 27 October 1961, in James M. Grimwood, Barton C. Hacker, and Peter J. Vorzimmer, *Project Gemini: A Chronology*, SP-4002 (Washington, D.C.: NASA, 1969), 14–15. Barton C. Hacker and James M. Grimwood, *On the Shoulders of Titans: A History of Project Gemini*, SP-4203 (Washington, D.C.: NASA, 1977), chaps. 2 and 3, 80–81.

27. Engineers originally planned for *Gemini* to land on solid ground using a "paraglider," similar to today's hang gliders.

28. Hacker and Grimwood, *On the Shoulders of Titans*, 82, 122.

29. Mary Louise Morse and Jean Kernahan Bays, *The Apollo Spacecraft, A Chronology*, vol. 2 (8 November 1962–30 September 1964), SP-4009 (Washington, D.C.: NASA, 1973), 70–71. Brooks et al., *Chariots for Apollo*, 121.

30. C. D. Stroud Briefing on Program Management Systems, "The Boeing Company Apollo/Saturn V Program Briefing, New Orleans, La., October 17, 1968," in *Apollo Program Management*, Staff Study for the Subcommittee on NASA Oversight of the Committee on Science and Astronautics, U.S. House of Representatives, 91st Congress, 1st Session, July 1969 (Washington, D.C.: Government Printing Office, 1969), 19–22.

31. Howard McCurdy, *Inside NASA* (Baltimore: Johns Hopkins University Press, 1993), 30–33.

32. C. D. Stroud, "S-IC Program Management Systems," in *Apollo Program Management*, 19–20. Bilstein, *Stages to Saturn*, 194–95. Lt. General Samuel Phillips, interview with Thomas W. Ray and Loyd S. Swenson Jr., 22 July 1970, JSC Archives Box 074-51, 27–28. Dunar and Waring, *Power to Explore*, 86–87.

33. Bilstein, *Stages to Saturn*, 277–78; quotation at 277. Brooks et al., *Chariots for Apollo*, 112–14.

34. Brooks et al., *Chariots for Apollo*, 119–20.

35. Lt. General Samuel Phillips, United States Air Force, interview with Tom Ray, 22 July 1970, NASA, "Phillips, Samuel C., "Interviews" folder 001701, LEK 1/13/4, NASAHO, 34–39. Morse and Bays, *Apollo Spacecraft*, vol. 2, 12, 76–77.

36. Brooks et al., *Chariots for Apollo*, 120–21.

37. See "Survey of Man-Rating Criteria," author unknown, n.d., ca. June–August 1964, "Notes Jun–Aug 1964," LC/SPP-43:9. Mitchell R. Sharpe and Bettye B. Burkhalter, "Mercury-Redstone: The First American Man-Rated Space Launch Vehicle," in John Becklake, ed., *History of Rocketry and Astronautics*, AAS History Series, vol. 17, R. Cargill Hall, series ed. (San Diego: AAS Publications Office, 1995), 341–88.

38. *Project Mercury Quarterly Status Report No. 6 for Period Ending April 30, 1960*, NASA, STG, NASAHO, 12. William B. Harwood, *Raise Heaven and Earth: The Story of Martin Marietta People and Their Pioneering Accomplishments* (New York: Simon and Schuster, 1993), 373–78. Swenson et al., *This New Ocean*, 254–56.

39. Owen G. Morris, JSC Oral History Project Interview with Summer Chick Bergen, 30 June 1999.

40. *Project Mercury Quarterly Status Report No. 6*, 12, 28. *Project Gemini Status Report No. 1 for Period Ending May 31, 1962*, NASA, MSC, NASAHO, 4, 5, 33–34. Harwood, *Raise Heaven and Earth*, 373–78. *Lockheed Agena Monthly Report, June 1964*, 3–11, in Grimwood et al., *Project Gemini*, 146. *Project Mercury Quarterly Status Report No. 7 for Period Ending July 31, 1960*, NASA, STG, NASAHO, 26.

41. Swenson et al., *This New Ocean*, 148–49, 270, and *Project Mercury Status Report No. 2 for Period Ending April 30, 1959*, NASA, Langley STG, NASAHO, 23–31. Brooks et al., *Chariots for Apollo*, 138. Morse and Bays, *Apollo Spacecraft*, vol. 2, 87. Harmon L. Brendle and James A. York, *Apollo Experience Report—The Command and Service Module Milestone Review Process*, NASA TN D-7599, February 1974, "Apollo Experience Reports" folder 007822, NASAHO, 1.

42. Bilstein, *Stages to Saturn*, chap. 4. John J. Williams and Donald M. Corcoran, "Mercury Spacecraft Pre-Launch Preparations at the Launch Site," paper presented at the American Institute of Aeronautics and Astronautics Space Flight Testing Conference, Cocoa Beach, Fla., 18–20 March 1963, 18. Swenson et al., *This New Ocean*, 180–82. McCurdy, *Inside NASA*, 32–33.

43. *Project Mercury Quarterly Status Report No. 6*, 5–11. *Project Mercury Quarterly Status Report No. 11 for Period Ending July 31, 1961*, NASA, MSC, NASAHO. *Project Mercury Quarterly Status Report No. 10 for Period Ending April 30, 1961*, NASA, STG, NASAHO.

44. Swenson et al., *This New Ocean*, 270–71.

45. *Project Mercury Quarterly Status Report No. 6*, 33–34. Grimwood et al., *Project Gemini*, 37, 68, 146. Morse and Bays, *Apollo Spacecraft*, vol. 2, 65.

46. Swenson et al., *This New Ocean*, 275–79, 308, 321–22.

47. Swenson et al., *This New Ocean*, 294–301. Dunar and Waring, *Power to Explore*, 81.

48. Memorandum from North to Director, Space Flight Programs, "Mercury Capsule Changes and Flight Schedule," in Swenson et al., *This New Ocean*, 299.

49. *Project Mercury Quarterly Status Report No. 15 for Period Ending July 31, 1962*, NASA, MSC, NASAHO, 26. *Project Mercury Quarterly Status Report No. 16 for Period Ending October 31, 1962*, NASA, MSC, NASAHO, 22.

50. Grimwood et al., *Project Gemini*, 94, 136, 164–65. *Gemini Program/Project Development Plan*, Gemini Program Office, M-D MG 600-1, 1 July 1966, NASA Office of Manned Space Flight, 7-4. Brooks et al., *Chariots for Apollo*, 123.

51. Jacob Neufeld, *Ballistic Missiles in the United States Air Force 1945–1960* (Washington, D.C.: Office of Air Force History, United States Air Force, 1990), 218.

52. Swenson et al., *This New Ocean*, 643. *Mercury Project Summary*, 4–5. Hacker and Grimwood, *On the Shoulders of Titans*, 96–115.

53. Hacker and Grimwood, *On the Shoulders of Titans*, 77.

54. W. Henry Lambright, *Powering Apollo: James E. Webb of NASA* (Baltimore: Johns Hopkins University Press, 1995), 114–17.

55. Hacker and Grimwood, *On the Shoulders of Titans*, 128–29, 581. *Project Gemini Status Report No. 7 for Period Ending November 30, 1963*, NASA, MSC, NASAHO, 84–85. Chamberlin's removal resulted from Seamans's embarrassment when he and McNamara visited MSC and found the program in far worse technical and budgetary shape than either were aware of. Cost increases were rampant in the *Titan II* launch vehicle, *Agena*, the paraglider, fuel cells, and the orbital maneuvering system. Originally cast as a minor modification of *Mercury*, *Gemini* ultimately used little of the *Mercury* design.

56. Lambright, *Powering Apollo*, 120–22. Levine, *Managing NASA*, 81.

57. *Preliminary Report to the Associate Administrator on Studies Relating to Management Effectiveness in Scheduling and Cost Estimating NASA Projects*, by Deputy Associate Administrator, 15 September 1964, NASAHO.

58. The DOD conversions to phased planning started based on the Bell Report of 1961. *Report to the President on Government Contracting for Research and Development*, Executive Office of the President, Bureau of the Budget, Washington, D.C., 30 April 1962 (Bell Report), in John M. Logsdon, ed., *Exploring the Unknown: Selected Documents in the History of the U.S. Civil Space Program*, SP-4218 (Washington, D.C.: NASA, 1996), 652–53. *Preliminary Report to the Associate Administrator.*

59. Levine, *Managing NASA*, 81, 158–60. *Apollo Program Weekly Status Report for Nov. 2–6, 1964*, from MAP-1/Chief, Program Planning, 6 November 1964, LC/SPP-41.

60. Biographical sketch of Mueller, in *Apollo Accident Hearings*, 8 parts, 90th Congress, 1st and 2d Sessions, 7 February 1967–January 1968, Senate Committee on Aeronautical and Space Sciences, Part 1, 68.

61. Interview with George E. Mueller, by Putnam, 27 June 1967, "George E. Mueller Interviews" folder 001522, LEK 1/12/1, NASAHO, 6–7.

62. Quotations from Mueller, interview by William P. Putnam, 12–13. Dunar and Waring, *Power to Explore*, 66.

63. Lambright, *Powering Apollo*, 117–18. Brooks et al., *Chariots for Apollo*, 129. Mueller, interview by Putnam, 8, 13–14.

64. Lambright, *Powering Apollo*, 117–18. Bilstein, *Stages to Saturn*, 348–50. Morse and Bays, *Apollo Spacecraft*, vol. 2, 104.

65. McCurdy, *Inside NASA*, 94–96. Bilstein, *Stages to Saturn*, 350–51. The "all-up" concept succeeded because von Braun's experienced team could anticipate most of the problems and plan tests accordingly.

66. "GEM" stood for George E. Mueller.

67. Logsdon, *Managing the Moon Program*, 19. Quotation from Owen Moriss.

68. *An Evaluation of Organization and Management, Office of Manned Space Flight, National Aeronautics and Space Administration*, prepared by Management Analysis Division, NASA headquarters, August 1964, "notes Jun–Aug 1964" LC/SPP-43:9, III-2.

69. Memorandum from George E. Mueller to NASA Administrator, 26 September 1963, Subject: Utilization of Air Force Program Management personnel, "Notes Jan-May" LC/SPP-43:8. Phillips, interview with Ray and Swenson, 41.

70. Jacob Neufeld, ed., *Reflections on Research and Development in the United States Air Force: An Interview with General Bernard A. Schriever and Generals Samuel C. Phillips, Robert T. Marsh, and James H. Doolittle, and Dr. Ivan A. Getting*, conducted by Dr. Richard H. Kohn (Washington, D.C.: Center for Air Force History, 1993), 87.

71. S. C. Phillips, Deputy Director, Apollo Program, letter to General B. A. Schriever, Commander AFSC, 20 January 1964, "NASA Personnel-Military" folder LC/SPP-43:15. W. E. Leonhard, Chief of Staff, AFSC, letter to Brig. Gen. S. C. Phillips, NASA, 14 February 1964, "NASA Personnel-Military" folder LC/SPP-43:15. Samuel C. Phillips, notes on fifty-five NASA positions to be filled by air force officers, ca. March 1964, "Notes, Jan-May" LC/SPP-43:8. Hugh L. Dryden, Deputy Administrator, NASA, letter to Eugene M. Zuckert, Secretary of the Air Force, 1 April 1964, "NASA Personnel-Military" folder LC/SPP-43:15. Eugene M. Zuckert, letter to Hugh L. Dryden, Deputy Administrator, NASA, 5 May 1964, "NASA Personnel-Military" folder LC/SPP-43:15. Hugh L. Dryden, letter to Eugene M. Zuckert, Secretary of the Air Force, 12 May 1964, "NASA Personnel-Military" folder LC/SPP-43:15. *Report of Joint Air Force/NASA Military Requirements Review Group*, September 1964, "NASA Personnel-Military" folder LC/SPP-43:15. Phillips, interview with Ray and Swenson, 15–16.

72. James E. Webb, letter to General J. McConnell, Vice Chief of Staff, USAF, 16 December 1964, "NASA 1964 Personnel-Military" LC/SPP-43:15. J. McConnell, General, USAF, Vice Chief of Staff, letter to Dr. Robert C. Seamans Jr., 21 December 1964, "NASA 1964 Personnel-Military" LC/SPP-43:15. Memorandum from David M. Jones (?), Brig. Gen., USAF, Deputy Associate Administrator for Manned Space Flight (Programs), to General McKee, Subject: Job descriptions for Colonels O'Connor, Bolender, and Yarchin, 15 January 1965, LC/SPP-43:15.

73. Memorandum for record from Samuel C. Phillips, 11 February 1964, "Chrono-

ogical File 1964" LC/SPP 41:2. Quotation from Samuel C. Phillips, notes, 29 February 1964, "Organization and Management, Conference in Los Angeles" LC/SPP-43:12.

74. Phillips, notes, 29 February 1964. Quotation from Samuel C. Phillips, notes and speech, ca. April 1964, "Organization and Management" LC/SPP-43:13.

75. Interoffice correspondence from C. W. Besserer, STL, to Brig. Gen. S. C. Phillips, Subject: Minuteman Control Room, 24 January 1963, "AF BSD Minuteman Correspondence 1963" LC/SPP-10:8. "Miscellaneous Viewgraphs," ca. January–February 1964, "Organization and Management" LC/SPP-43:11.

76. Bilstein, *Stages to Saturn*, 284–86. Tompkins, *Organizational Communication Imperatives*, 78–79. Dunar and Waring, *Power to Explore*, 67–68.

77. *Apollo Configuration Management Manual*, NPC 500-1, Office of Manned Space Flight, Apollo Program, NASA, 18 May 1964, "Config Mgmt Manual" LC/SPP-41:3. *Apollo Executives' Meeting Proceedings*, Asheville, N.C., 18–19 June 1964, "Executives Meeting" LC/SPP-42:6, III-8. Memorandum from Samuel C. Phillips to Marshall Space Flight Center, Apollo Program Manager, Subject: Apollo Configuration Management Manual, 25 May 1964, "Config Mgmt Manual" LC/SPP-41:3. R. B. Young, Director, Industrial Operations, MSFC, Letter to Maj. General Samuel C. Phillips, Subject: Apollo Configuration Management Manual, 19 June 1964, "Config Mgmt Manual" LC/SPP-41:3.

78. Memorandum to distribution from E. L. Harkleroad, Subject: Configuration management—development of Apollo specifications, 20 July 1964, LC/SPP-43:9.

79. *Configuration Management Provisional Group Report, SID 64-1608*, 15 September 1964, North American Aviation, Space and Information Systems Division, LC/SPP-41, 1, 25, 59–63; quotation at 25.

80. Col. John V. Patterson Jr., letter to Brig. Gen. S. C. Phillips, Director, Minuteman SPO, AFSC/BSD, 7 April 1963, LC/SPP-37. Memorandum from Maj. General Samuel C. Phillips, Deputy Director, Apollo Program, to MSFC, KSC, MSC, Subject: System management course at Air Force Institute of Technology, 10 October 1964, LC/SPP-37. See also memorandum from Maj. General Samuel C. Phillips to Admiral Boone, Subject: Defense Weapon Systems School, 1 December 1964, LC/SPP-41. This memorandum shows that two people from MSFC and two from HQ went to the class.

81. *Apollo Executives' Meeting Proceedings*, III-11, 13.

82. Ibid., III-16, 17.

83. Ibid., III-18.

84. Ibid., III-19.

85. Ibid., III-20.

86. Memorandum to file by Stanley Winn and Lt. Col. C. Taylor, Subject: Trip Report to MSC, Houston, on October 6 and 7, 1964, "Notes Sept-Dec 1964" LC/SPP-43:10. Quotation from Samuel C. Phillips, "Configuration Control," speech at Southeastern Symposium on Government Contracts, Cocoa Beach, Fla., 19 March 1965, LC/SPP-55:8, 1. Lee B. James, Manager, Saturn I/IB Program, MSFC, 26 August 1965, "Letter to Maj. General Samuel C. Phillips" LC/SPP-55:3.

87. "Common Industry Problems Found among Apollo Contractors," n.d., ca. mid-1965, "Systems Management" LC/SPP-55:3.

88. *NASA-Apollo Program Management*, vol. 1 (Washington, D.C.: NASA OMSF Apollo Program Office, December 1967), NASAHO, 4-44–4-46. *Apollo Program Directive No. 6A, Subject: Sequence and Flow of Hardware Development and Key Inspection, Review and Certification Checkpoints*, 30 August 1966, in *NASA-Apollo Program Management*, vol. 1, December 1967, app. J.

89. Courtney G. Brooks and Ivan D. Ertel, *The Apollo Spacecraft, A Chronology*, vol. 3, 1 October 1964–20 January 1966, SP-4009 (Washington, D.C.: NASA, 1976), 51, 97.

90. *NASA-Apollo Program Management*, vol. 1, 4-28–4-34.

91. Mike Gray, *Angle of Attack: Harrison Storms and the Race to the Moon* (New York: Penguin, 1992), 196–98. Dunar and Waring, *Power to Explore*, 89–90.

92. Bilstein, *Stages to Saturn*, 224–25.

93. "Tiger team" was an Air Force term for an ad hoc group to dig up and fix critical problems using the best available personnel. Bilstein, *Stages to Saturn*, 225. Phillips, interview with Ray and Swenson, 29–30.

94. Phillips to Mueller, quoted in Lambright, *Powering Apollo*, 152. Gray, *Angle of Attack*, 200–202.

95. Gray, *Angle of Attack*, 207–9. Bilstein, *Stages to Saturn*, 227–29. "Grumman Aircraft Engineering Corporation, Summary of the Report for the Subcommittee on NASA Oversight, November 4, 1968," in *Apollo Program Management*, House Staff Study, 45. See also "North American Rockwell Corporation," in *Apollo Program Management*, 175, 188. Harvard Business School professor J. Sterling Livingston knew about work package management in 1962. However, it did not reach NASA until North American accelerated its deployment at this time. See J. Sterling Livingston, Harvard University, "Advanced Techniques for Program/Funds Management," in *AFSC Management Conference*, Monterey, Calif., 2–4 May 1962 (Washington, D.C.: Air Force Systems Command, 1962).

96. "North American Rockwell Corporation," 174, 189.

97. Brooks et al., *Chariots for Apollo*, 197–201. Ivan D. Ertel and Roland W. Newkirk, with Courtney G. Brooks, *The Apollo Spacecraft: A Chronology*, vol. 4, 21 January 1966–13 July 1974, SP-4009 (Washington, D.C.: NASA, 1978), 27.

98. *Report of Apollo 204 Review Board to the Administrator*, NASA, 1967, by Floyd L. Thompson, chairman, 5 April 1967, with Appendices A–G, NASAHO. Gray, *Angle of Attack*, 226–31.

99. See Lambright, *Powering Apollo*, chaps. 8 and 9, for an excellent description of Webb's role in the aftermath of the fire.

100. *Apollo Accident Hearings*, 233. Gray, *Angle of Attack*, 232–35.

101. Lambright, *Powering Apollo*, 151–58.

102. Memorandum from James E. Webb, Administrator NASA, for Mr. Shapley/ADA, Subject: Problem of adding "supervision" to the activities of top management

to see that the plans made are carried out, both in the area of substance and in the area of administration, 19 September 1967, NASAHO.

103. Lambright, *Powering Apollo*, 161. James E. Webb, Administrator NASA, memorandum to Mr. Finger, 27 October 1967, NASAHO.

104. Lambright, *Powering Apollo*, 170–75. Gray, *Angle of Attack*, 244–56.

105. Lambright, *Powering Apollo*, 175–76. Phillips, interview with Tom Ray, 39–41. Phillips knew Boeing well because it was the Minuteman contractor.

106. See "The Boeing Company Apollo/Saturn V Program Briefing," in *Apollo Program Management*, House Staff Study, 31–34.

107. "Single point failure" is a term meaning any single fault that could cause a loss of the mission or loss of life. In general, it is impossible to eliminate all single-point failures from a design (the spacecraft structure is a typical example). Therefore, NASA found it necessary to identify each one and ensure that the probability of its occurrence was very small.

108. Ertel and Newkirk, *Apollo Spacecraft*, vol. 4, 152–53, 163, 176–78, 198. Memorandum from Sam C. Phillips to Mueller, 5 September 1968, George E. Mueller Correspondence folder 001553, LEK 1/12/1, NASAHO. This list is not exhaustive.

109. "North American Rockwell Corporation," 150–54.

110. Ertel and Newkirk, *Apollo Spacecraft*, vol. 4, 203–5, 223–24. Letter from George E. Mueller, NASA OMSF, to Robert R. Gilruth, MSC, 14 February 1968, NASAHO. Memorandum from Mueller to Phillips, Subject: Software Task Force meetings, 19 February 1968, NASAHO.

111. Ertel and Newkirk, *Apollo Spacecraft*, vol. 4, 250, 257.

112. Dunar and Waring, *Power to Explore*, chap. 5.

113. Tompkins, *Organizational Communication Imperatives*, 105–11.

114. Ibid., 112.

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R. Aubinière, "Tenth Anniversary of the Establishment of ELDO," *ESRO/ELDO Bulletin* 24 (March 1974): 13.

1. Alfred Grosser, *The Western Alliance: European American Relations since 1945*, trans. Michael Shaw (New York: Vintage Books, 1982), from *Les Occidentaux, Les pays d'Europe et les Etats-Unis depuis la guerre* (Paris: Editions Fayard, 1978). Alan S. Milward, *The Reconstruction of Western Europe 1945–51* (Berkeley: University of California Press, 1984), and *The European Rescue of the Nation-State* (Berkeley: University of California Press, 1992). Charles Maier, *In Search of Stability* (Cambridge: Cambridge University Press, 1987).

2. Armin Hermann et al., *History of CERN* (Amsterdam: North-Holland Physics Publishing, 1987), 2 vols. Jaroslav G. Polach, *EURATOM* (Dobbs Ferry, N.Y.: Oceana Publications, 1964).

3. Jean-Jacques Servan-Schreiber, *The American Challenge*, trans. Robert Steel (New York: Avon Books, 1969), 168 (emphasis in original).